

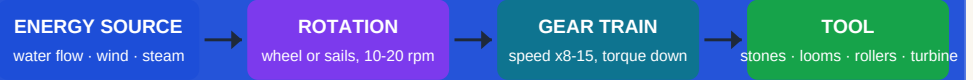
THE MILL

From Water Wheels to Wind Turbines

Watermills · Windmills · Textile Mills · Modern Mills - full history + how they work

WHAT IS A MILL? — THE 2,000-YEAR MACHINE

A **mill** is a machine that captures energy - from flowing water, wind, muscle, steam or electricity - and converts it into **rotary motion** that drives tools: grindstones, hammers, saws, rollers, bellows and looms. The word comes from Latin ***mola** *, meaning millstone. Every mill, from a Roman water wheel to a modern wind turbine, runs the same four-step chain:



Types of mill by job: grain/flour mills, fulling mills (cloth), paper mills, sawmills, textile mills (cotton & wool), steel rolling mills, ore-crushing stamp mills, gunpowder mills, oil mills, and today's wind turbines & hydroelectric plants. One word - one universal machine.

Why mills matter

Mills were the **first machines in history** to use gears, bearings and power transmission - they invented mechanical engineering 2,000 years before engines existed. Windmills and waterwheels were humanity's only engine until the steam engine, and mill mechanisms (cams, governors, gear trains) were copied directly into the first factories.

TIMELINE — 2,200 YEARS OF MILLS

Year	Milestone
~200 BCE	Rotary quern hand-mills spread across Europe - two stones, one turned by a handle.
1st c. BCE	Vitruvius describes the geared vertical waterwheel (undershot) - gears appear in a machine for the first time.
~300 CE	Barbegal, France : 16 overshot wheels in 2 rows power Rome's biggest industrial flour plant (~25 t/day).
~619	First windmills recorded in Persia (Sistan) - vertical-axis, panemone design.
1086	Domesday Book counts 5,624 watermills in England alone - ~1 per 350 people.
1185	First recorded windmill in England (Yorkshire).
12th c.	Post mills appear - the whole body pivots on a central post to face the wind.
14th c.	Tower mills (fixed stone tower, rotating cap) in France/UK; smock mills follow in the Low Countries.
1550s	Cam-driven trip hammers & metal gears turn mills into proto-factories (ore crushing, fulling).
1745	Edmund Lee's fantail - the first automatic control device: a small wind wheel that steers the cap into the wind.
1764	Hargreaves' spinning jenny - 8+ spindles per spinner.
1769	Arkwright's water frame - water-powered spinning; the first true factory machine.
1779	Crompton's spinning mule - up to 1,000+ spindles.
1785	Cartwright's power loom ; steam mills soon replace water in Manchester ('Cottonopolis').
1807	Fourdrinier paper-making machine - continuous paper rolls.
1870s	Roller mills replace millstones for flour (Stevens process) - cleaner, faster, higher extraction.
1887	First wind turbine for electricity (Blyth, Scotland).
1895	First large hydroelectric plant (Niagara Falls).
Today	3-blade HAWT wind turbines & Francis/Kaplan hydro turbines; roller mills make >95% of world flour.

WATERMILLS — HOW THEY WORK

Water flows onto the wheel's blades or buckets, the wheel turns (typically **10-20 rpm**), and a gear train multiplies the speed up to millstone speed (**100-150 rpm**). The classic drive chain:

Water wheel on the axle turns the **pit wheel** (large vertical gear).

Pit wheel drives the **wallower** - a small lantern gear on a horizontal shaft (first speed step-up, ~2-4x).

The wallower's shaft drives the **great spur wheel** (horizontal, like a crown gear).

A **stone nut** (pinion) on each millstone meshes with the spur wheel - the stone spins at 100-150 rpm.

Grain drops from the **hopper** through the shoe into the **eye** (centre hole) of the rotating **runner stone** .

The runner grinds grain against the fixed **bed stone** ; flour is flung out at the rim and falls down the meal spout.

Wheel types (efficiency = how well water energy is captured)

Type	How water hits the wheel	Efficiency
Undershot	Flat blades dip into the stream below - simplest, works in low-head/high-flow rivers	20-30%
Breastshot	Water meets the wheel at axle height (mid-level)	50-70%
Pitchback	Water enters at the top but turns the wheel backwards (self-cleaning buckets)	~75%
Overshot	Buckets fill at the top from a flume/launder - gravity does the work, needs ~2 m+ head	70-85%
Poncelet	Curved blades, undershot position, redirects flow for extra push	50-70%

Power of a waterwheel: $P = \text{density} \times g \times \text{flow} \times \text{head} \times \text{efficiency}$, i.e. **$P \text{ (watts)} = 1000 \times 9.81 \times Q \text{ (m}^3/\text{s)} \times h \text{ (m)} \times \text{efficiency}$** . Worked example: $Q = 0.5 \text{ m}^3/\text{s}$, $h = 3 \text{ m}$, 75% efficient -> $P = 1000 \times 9.81 \times 0.5 \times 3 \times 0.75 \sim 11,000 \text{ W} \sim 15 \text{ hp}$ - enough for 4 pairs of stones (one pair needs 3-10 hp).

Beyond flour: fulling cloth (trip hammers), sawing timber, crushing ore (stamp mills), pumping water (Archimedes screw), blowing furnace bellows, and powering Arkwright's cotton mills (Cromford, 1771).

FUN FACTS

The **Domesday Book** (1086) counted 5,624 watermills in England - more mills per person than most of Europe.

Barbegal (~300 CE): Rome's 16-wheel industrial estate ground ~25 tonnes of flour a day for Aries.

The **first windmills** (Persia, 7th c.) spun on a **vertical axis** - a completely different design from today's.

Mill idioms everywhere: **'run of the mill'**, **'grist to the mill'**, **'millstone around your neck'** .

The **mill governor** was the first feedback device (ancestor of the thermostat) - Watt copied it for steam engines, and 'milling' a coin's edge stopped silver-clipping.

Windmills drained the sea and created ~ **17% of the Netherlands'** current land area.

WINDMILLS — HOW THEY WORK

Wind pushes the **sails** (usually 4, radius 8-10 m), the **windshaft** turns at 10-20 rpm, and the same gear train as a watermill steps the speed up to the stones:

Sails -> **windshaft** -> **brake wheel** (gear on the brake rim).

Brake wheel drives the **wallower** -> **upright shaft** (vertical).

Upright shaft's **great spur wheel** drives the **stone nuts** -> millstones at 100-150 rpm.

The **brake** is a friction band around the brake wheel - the windmill's 'brake pedal', pulled from inside.

Keeping the sails into the wind: early mills were turned by hand with a tailpole. From 1745 the **fantail** - a small wind wheel mounted at right angles on the back - automatically rotates the **cap** (the roof) on its track through worm gears. The fantail is the **first feedback control device in history** - the ancestor of the thermostat.

Types & sail designs

Post mill (12th c.): the whole wooden body pivots on a central post - oldest design.

Tower mill (14th c.): brick/stone tower, only the cap rotates - stronger, taller, sails up to 30 m.

Smock mill (16th c.): wooden body shaped like a smock, cheaper than stone - common in NL & UK.

Common sails : canvas sheets reefed (shortened) by hand; **Spring sails** (1772): self-setting shutters; **Patent sails** (1807): shutters adjusted while turning; **Roller reefing** (1950s): fabric rolled up like a blind.

Power of a windmill: $P = 0.5 \times \text{air density} \times \text{swept area} \times \text{wind speed}^3 \times C_p$. Air density ~1.225 kg/m3, area = $\pi \times r^2$, and C_p (rotor coefficient) is at most the **Betz limit 0.593** - no rotor can capture more than 59.3% of wind energy. Worked example: $r = 10 \text{ m}$, $v = 8 \text{ m/s}$, $C_p = 0.4 \rightarrow P = 0.5 \times 1.225 \times 314 \times 512 \times 0.4 \sim 39,400 \text{ W} \sim 53 \text{ hp}$. Note the v^3 : doubling wind speed gives 8x the power - location is everything.

Famous jobs: draining Dutch **polders** (land reclaimed below sea level, pumped by wind-driven Archimedes screws - ~17% of the Netherlands was won from the sea), grinding grain, sawing timber, pressing oil and paper-making.

GEARS & MECHANISMS — THE ENGINEERING CORE

Mills forced engineers to invent **power transmission** - here are the components, every one still used in modern machinery:

Crown gear : flat ring with teeth on its face - lets a vertical shaft drive a horizontal one (Roman, 1st c. BCE).

Bevel gear : teeth cut on cone faces - changes axis direction at any angle.

Lantern gear / wallower : wooden staves between two discs - cheap, forgiving, self-cleaning; the most common mill gear.

Spur gear : teeth on the rim, parallel shafts - the workhorse of every machine since.

Worm gear : a screw driving a wheel - huge reduction, used in fantails and cap tracks.

Cam : pegs on a rotating shaft that lift and drop **trip hammers** - turns rotation into reciprocating hammering (fulling, ore stamping).

Governor : spinning weights that sense speed and adjust sail shutters or the gap between millstones - the first automatic speed control.

Archimedes screw : helix inside a tube that lifts water - invented ~250 BCE, still used in sewage plants today.

Gearing maths: speed ratio = teeth on driven gear / teeth on driver. A 48-tooth pit wheel driving an 8-stave wallower steps speed up 6x; a spur wheel of 60 teeth driving a 15-tooth stone nut steps up 4x more. Chain: 12 rpm wheel x 6 x 2.5 ~ 180 rpm at the stone. Gear teeth were cut from wood and iron long before precision metal-cutting existed - millwrights were the first mechanical engineers.

TEXTILE MILLS — THE INDUSTRIAL REVOLUTION

Cotton demand exploded in the 1700s, but spinning was the bottleneck: one spinner fed only a fraction of one weaver, so yarn was scarce and costly. A generation of machines solved it - and created the **factory system** :

Spinning jenny (Hargreaves, 1764): hand-cranked frame with 8-120 spindles - one worker spins many threads at once.

Water frame (Arkwright, 1769): powered rollers stretch the roving while spindles twist it - strong warp yarn, and water-powered: a machine too big for a cottage, so workers came to it.

Spinning mule (Crompton, 1779): carriage + rollers + spindles combined - fine AND strong yarn; powered mules with 1,000+ spindles by 1800.

Power loom (Cartwright, 1785): automated weaving to match the spinning machines.

Factory system : Arkwright's Cromford Mill (1771, Derwent Valley) - 24-hour shifts, child 'parish apprentices', purpose-built mill village with housing - the template for every factory since.

Steam : Boulton & Watt engines (1780s) freed mills from rivers; Manchester became 'Cottonopolis' with 100+ mills.

Human cost: 12-16-hour days, unguarded machinery, byssinosis ('brown lung') from cotton dust. **Factory Act 1833** : no children under 9, 9-13s limited to 9 h/day; **1847 Ten Hours Act** capped women & young people - the birth of labour law.

Impact: cotton exports multiplied ~100x in a century; canals, railways and mill towns followed; capital investment and wage labour replaced cottage work. The mill gave the world the **modern factory, shift work and mass production** .

MODERN MILLS — SAME CHAIN, NEW POWER

The capture -> rotate -> gear -> tool chain survives in every modern industry:

Flour - roller mills (1870s): fluted steel **break rolls** crack the grain, sifters separate bran/germ/endosperm, smooth **reduction rolls** grind to flour; continuous, hygienic, ~75% extraction - >95% of world flour today.

Paper : the **Fourdrinier machine** (1807) - pulp slurry onto a moving wire mesh, water drains, press rollers squeeze, heated drying cylinders finish - paper rolls at 1,500+ m/min.

Steel : **rolling mills** squeeze red-hot ingots through successive roller stands (hot rolling), then cold rolling for precision - sheets, rails, beams for the modern world.

Sawmills : band saws & circular saws, laser scanning, debarking, chipping - computerised from log to plank.

Textiles today : automated ring spinning and air-jet looms (1,000+ picks/min) - a mule shed that needed 500 workers now runs with 10.

Hydroelectric : dammed 'head' drives Pelton (high-head), Francis (medium), Kaplan (low-head) turbines - ~16% of world electricity.

Wind turbines : 3-blade HAWT, 100+ m rotor, gearbox -> generator, still Betz-limited like their ancestors - ~7% of world electricity and climbing.

2,200 YEARS OF HISTORY

GEARS, CAMS & GOVERNORS

POWER FORMULAS INSIDE

MILL MATH — QUICK REFERENCE

Water power

$P \text{ (W)} = 1000 \times 9.81 \times Q \times h \times \text{efficiency}$ (Q = flow m3/s, h = head m)■Example: $0.5 \text{ m}^3/\text{s} \times 3 \text{ m} \times 75\% \rightarrow \sim 11 \text{ kW} \sim 15 \text{ hp}$

Wind power

$P = 0.5 \times 1.225 \times A \times v^3 \times C_p$ ($A = \pi r^2$, v = wind speed m/s)■Example: $r = 10 \text{ m}$, $v = 8 \text{ m/s}$, $C_p = 0.4 \rightarrow \sim 39 \text{ kW}$ ■Betz limit: $C_p \text{ max} = 0.593$

Gearing

ratio = driven teeth / driver teeth ■wheel 12 rpm x 6 (pit/wallower) x 2.5 (spur/stone nut) ~ 180 rpm

Units & rules of thumb

1 hp = 746 W = 550 ft-lb/s (Watt's horse)■One pair of stones: 3-10 hp; wheel 10-20 rpm -> stones 100-150 rpm■Wind power scales with v^3 (2x wind = 8x power)

GLOSSARY — 28 KEY TERMS

Quern	Hand-milltwo stones, upperturnedby handle	Fantail	Auto-steeringwind wheel(1745)
Runner stone	Rotating upper millstone	Common / patent sails	Canvas vs shutter-type sails
Bed stone	Fixed lower millstone	Post / tower / smock	Pivoting body / stone tower / wooden body
Eye	Centre hole where grain enters	Undershot / overshot	Water hits below / top of the wheel
Hopper & shoe	Grain funnel + feed chute to the eye	Breastshot	Water hits wheel at axle height
Pit wheel	Large vertical gear on wheel axle	Head	Vertical drop of water available
Wallower	Small lantern gear (first step-up)	Flume / launder	Channel carrying water to the wheel
Great spur wheel	Horizontal gear driving stone nuts	Fulling	Beating wool to felt it (trip hammers)
Stone nut	Pinion driving one millstone	Trip hammer / cam	Lifted-and-dropped hammer on cams
Lantern gear	Cylindrical stave gear - cheap & forgiving	Governor	Auto speed control - first feedback device
Crown gear	Flat ring gear with face teeth (Roman)	Break / reduction rolls	Fluted crack rolls vs smooth flour rolls
Windshaft	Main axle carrying the sails	Betz limit	Max 59.3% of wind energy capturable
Brake wheel	Gear on windshaft; also the brake drum	HAWT	Horizontal-Axis Wind Turbine
Cap	Rotating roof of tower/smock mills	Pelton / Francis / Kaplan	High / medium / low-head turbines

QUICK-FIRE QUIZ — 12 REVISION QUESTIONS

#	Question	Answer
1	Which water wheel type is most efficient?	Overshot (70-85%) - gravity does the work
2	What does the wallower do?	Small lantern gear stepping speed up from the pit wheel
3	Who invented the water frame (1769)?	Richard Arkwright - first factory machine
4	What is the Betz limit?	Max 59.3% of wind energy any rotor can capture
5	What did the fantail (1745) automate?	Turning the cap into the wind - first feedback device
6	Name the 4 windmill sail types	Common (canvas), spring, patent, roller-reefing
7	Which gear design is the oldest?	Crown gear - described by Vitruvius, 1st c. BCE
8	What drives a trip hammer?	Cams on a rotating shaft (fulling & ore crushing)
9	How fast do millstones spin?	100-150 rpm (wheel itself: 10-20 rpm)
10	1 horsepower = ?	746 W (550 ft-lb/s - Watt's measure)
11	What drained the Dutch polders?	Windmills driving Archimedes screws (~17% of NL)
12	What replaced millstones for flour?	Roller mills, 1870s (Stevens process)